C01 Sprites and Sorting

CRE132 worksheet

What you'll make

A small 2D scene built from pixel-art sprites: a grass floor, a tree, the player, a bush and a little pile of rocks. There is no code in this sample; everything happens in the Unity editor.

You'll learn how Unity turns a picture into a **sprite**, how one picture holding many drawings (a *sprite sheet*) is cut up, which import settings keep pixel art sharp, and how Unity decides which sprite draws in front of which: **Sorting Layers** and **Order in Layer**.



Figure 1: The finished scene: the player stands in front of the tree but behind the bush, and three rocks overlap in a set order.

Before you start

1. Unzip `CRE132_C01_SpritesAndSorting_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_C01_SpritesAndSorting_Starter` folder. Open it with Unity **6000.3.11f1**. The first time, Unity takes a minute or two to import everything.
2. In the **Project** window (bottom of the screen), open the folder `Assets/_game/Scenes/Core` and double-click the scene `C01_SpritesAndSorting`.
3. Press **Play** (the triangle at the top). The Game view is plain dark green: the scene has no sprites yet. Press Play again to stop.

Find your way around the main windows:

- **Hierarchy** (left): every object in the scene. Here there are only two: **Main Camera** and **Global Light 2D**. You'll add everything else.
- **Scene** view (middle): where you build. Scroll to zoom, hold the right mouse button (or middle button) and drag to pan. Click an object to select it.

- **Game** view: what the camera sees when you press Play.
- **Inspector** (right): the settings of whatever you selected.
- **Project** window: the files in the project. The art is in `Assets/_game/Art/Sprites/MysticWoods`, already imported for you.

Save often with Ctrl+S.

1. Import settings for pixel art

In the Project window, open `Assets/_game/Art/Sprites/MysticWoods` and click `player.png`. The Inspector shows its **import settings**, which tell Unity how to turn the picture file into sprites:

- **Texture Type: Sprite (2D and UI).** Makes it a sprite you can put in a 2D scene.
- **Sprite Mode: Multiple.** This one picture holds many drawings (every frame of the player's animations).
- **Pixels Per Unit: 16.** 16 pixels of art fill one Unity unit, so one 16×16 tile is exactly one square of the grid.
- **Filter Mode: Point (no filter).** Each art pixel stays a sharp square.
- **Compression: None.** Compression smudges the colours of small pixel art.

See why Filter Mode matters. First you need a sprite in the scene: click the small arrow on `player.png` to open it, and drag `player_0` into the Scene view. A small player appears; scroll to zoom in on it. Now click `player.png` again, change Filter Mode to **Bilinear** and click **Apply**. The player's edges go blurry, because Unity blends neighbouring pixels. Set it back to **Point (no filter)** and click **Apply**.

`objects.png` and `grass.png` use the same settings. Click them to check.

2. Slicing a sprite sheet

Select `player.png` and click **Open Sprite Editor** in the Inspector. The sheet has already been cut into 48×48 pixel frames. You can see how in the **Slice** menu: **Type: Grid By Cell Size**, **Pixel Size** 48 × 48. Each row is one animation: rows 0 to 2 are idle, 3 to 5 moving, 6 to 8 attacking and row 9 is death. Close the Sprite Editor.

Now open the Sprite Editor for `objects.png`. Here the drawings are all different sizes, so a grid won't work. Try **Slice > Type: Automatic** and click **Slice**: Unity draws a box round each separate drawing, but drawings that touch get merged into one. **Don't click Apply.** Click **Revert** to get the original boxes back.

The boxes you get back were drawn by hand and named (**tree**, **bush**, **rock_1**, **rock_2**...). Click one: its **Pivot** is **Bottom Center**, so the object stands on its base. That matters later for depth sorting (T03). Close the Sprite Editor.

In the Project window, click the small arrow on `objects.png`: every sprite is listed inside it.

3. Placing sprites

You already have `player_0` in the scene. Drag these sprites from the Project window into the Scene view (or the Hierarchy):

- `grass` (from `grass.png`)
- `tree` and `bush` (inside `objects.png`)
- `rock_1`, `rock_2` and `rock_3` (inside `objects.png`)

Each one becomes a GameObject with a **Sprite Renderer**, the component that draws a sprite. Rename them in the Hierarchy (select, press F2): **Ground**, **Tree**, **Bush**, **Player**, **Rock (Order 0)**, **Rock (Order 1)** and **Rock (Order 2)**.

Move them with the **Move** tool (press W and drag the arrows), or type the **Position** in the Inspector:

Object	Position X	Position Y
Tree	−4.5	−2.5
Player	−4.71	−1.24
Bush	−3.8	−3.5
Rock (Order 0)	3	−1.5
Rock (Order 1)	3.6	−1.3
Rock (Order 2)	4.2	−1.5

Test it: you probably can't see some of them, or they overlap the wrong way. The next two parts fix that.

4. A tiled grass floor

Select **Ground**. In its Sprite Renderer, set **Draw Mode** to **Tiled** and **Size** to **20** × **12**. Set its **Position** to 0, 0.

- **Tiled** repeats the 16×16 grass sprite to fill the size you give, instead of stretching it. 20 × 12 units is bigger than the camera's view, so the grass fills the screen.
- Tiling only works when the sprite's **Mesh Type** is **Full Rect**. Click `grass.png` in the Project window to see that it is.

Test it: the Game view is now green from edge to edge. But the grass may be covering the other sprites.

5. Sorting Layers

Select **Ground**. In the Sprite Renderer, click the **Sorting Layer** drop-down and choose **Add Sorting Layer...**. The **Tags and Layers** window opens with this project's sorting layers:

1. Default
2. Background
3. Objects
4. Characters
5. Foreground

They've already been made for you. **Lower in the list draws on top**, so Foreground is drawn last, over everything else. Sorting layers belong to the whole project, so every later sample reuses these.

Add it: set the **Sorting Layer** of each object:

Object	Sorting Layer
Ground	Background
Tree	Objects
Player	Characters
Bush	Foreground
the three rocks	Objects

Test it: drag the **Player** around over the tree and the bush. The player always draws in front of the tree and behind the bush, wherever you put it. The layer wins over the position.

6. Order in Layer

The three rocks are all on **Objects**, so the sorting layer can't decide between them. **Order in Layer** does: on the same layer, higher numbers draw on top.

Add it: set **Order in Layer** to **0** on **Rock (Order 0)**, **1** on **Rock (Order 1)** and **2** on **Rock (Order 2)**.

Test it: the rocks overlap with Rock (Order 2) at the front. Swap two of the numbers and watch the pile change.

Finally, tidy the Hierarchy: right-click, **Create Empty**, rename it **Sorting Layers Demo** and drag **Tree**, **Player** and **Bush** onto it. Do the same for the rocks with an empty called **Order In Layer Demo**. Grouping doesn't change how anything draws; it just keeps the Hierarchy readable.

Check your work

- ☐ The pixel art is sharp, not blurry.
- ☐ The grass fills the Game view.
- ☐ Wherever you drag the player, it draws in front of the tree and behind the bush.
- ☐ The three rocks overlap in order 0, 1, 2, with 2 at the front.
- ☐ The Hierarchy has **Ground**, **Sorting Layers Demo** (Tree, Player, Bush) and **Order In Layer Demo** (the three rocks), and the scene is saved.
- ☐ Compare with `CRE132_C01_SpritesAndSorting_Finished` if something looks different.

Key points

- **Pixel art settings:** Pixels Per Unit to match the tile size (16 here), Filter Mode **Point**, Compression **None**.
- **Sorting Layer first, then Order in Layer.** Unity picks the layer first (lower in the list draws on top), then, on the same layer, the higher Order in Layer draws on top.
- **Position doesn't decide** which sprite is in front when they are on different sorting layers, not even the Z position.
- **Sorting layers are project-wide.** Every scene in the project, and every later sample, uses the same list.

If something goes wrong

- **A sprite I dragged in is invisible:** it's probably behind the grass. Check its Sorting Layer, and that Ground is on **Background**.
- **The grass is stretched, not repeated:** Draw Mode is still **Simple**. Set it to **Tiled**.
- **The art is blurry:** a texture's Filter Mode is not **Point (no filter)**, or you forgot to click **Apply**.
- **I clicked Apply after Automatic slicing and the sprites changed:** press Ctrl+Z, or close the project without saving and unzip a fresh copy.
- **I can't find the Sorting Layer field:** select the object in the Hierarchy and look in the **Sprite Renderer** component, under **Additional Settings**.

Try it

1. Move the **Bush** to the **Objects** layer. Now drag the player past it: what changes?
2. Give two rocks the same Order in Layer and move them around. Which one ends up in front?
3. Change the Ground's **Size** to 10 × 6 and press Play. What does the camera show past the edge?